

Systems of irrational equations

Four lessons on the systems that hide a quadratic — sum and difference of roots, substitution, and the conditions that must survive to the end.

LESSON 43–46

4 × 40 MIN

ALGEBRA 10, §2.6

P1 · 1.5–1.7 (EXTENSION)

LESSON CLOCK

16	30'	30'	30'	50'	24'
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Where the difficulty is 16 min

$u = \sqrt{x}$, $v = \sqrt{y}$ 30 min

Sum and product of roots 30 min

Mixed systems 30 min

Practice 50 min

Homework and consolidation 24 min

LEARNING OBJECTIVES

- Solve a system of irrational equations by substitution.
- Use $u = \sqrt{x}$, $v = \sqrt{y}$ to turn a system into a polynomial one.
- Track the non-negativity conditions through every step.
- Check every candidate pair in the original system.

TEXTBOOK

National pp. 163–180

Cambridge Pure Mathematics 1, pp. 12–23

Workbook P1 Exercise 1E, 1F

01 Explanation

Where the difficulty is

Nothing new is being asked. A system of irrational equations combines two techniques you already have — the substitution of the last lesson and the elimination of Lesson 30–31. What is new is the **bookkeeping**: two non-negativity conditions instead of one, and they must survive every step.

WRITE THE CONDITIONS ONCE, AT THE TOP

$x \geq 0$, $y \geq 0$, and any others the radicands impose. Then substitute $u = \sqrt{x} \geq 0$ and $v = \sqrt{y} \geq 0$ — and every rejection later is decided by those two inequalities alone.

The standard substitution

$$\sqrt{x} + \sqrt{y} = 5 \text{ and } x + y = 13$$

Put $u = \sqrt{x}$, $v = \sqrt{y}$, both ≥ 0 . Then $x = u^2$ and $y = v^2$, and the system becomes polynomial:

$$u + v = 5 \text{ and } u^2 + v^2 = 13$$

So $uv = \frac{25 - 13}{2} = 6$, and u, v are the roots of $t^2 - 5t + 6 = 0$: 2 and 3. Both are non-negative, so both are kept.

$(x, y) = (4, 9) \text{ or } (9, 4)$

SQUARE THE SUBSTITUTION BACK CAREFULLY

$u = 2$ gives $x = 4$, not $x = 2$. Half the errors in this lesson are made in the last line.

Sum and product of the roots

Many systems are stated directly in $\sqrt{x} + \sqrt{y}$ and $\sqrt{xy}$. Since $\sqrt{xy} = \sqrt{x} \cdot \sqrt{y}$ for non-negative x, y , these are exactly $u + v$ and uv :

GIVEN	IN u, v
$\sqrt{x} + \sqrt{y}$	$u + v = s$
$\sqrt{xy}$	$uv = p$
$x + y$	$s^2 - 2p$
$x - y$	$(u - v)(u + v)$

Then u and v are the roots of $t^2 - st + p = 0$, and both must come out non-negative or the pair is rejected.

Mixed systems

When only one equation carries a radical, isolate and substitute in the ordinary way:

$\sqrt{x + y} = 3 \text{ and } x - y = 1$

The first gives $x + y = 9$ (and requires $x + y \geq 0$, which it now is), so the system is linear: $x = 5, y = 4$. Check: $\sqrt{9} = 3$ ✓.

THE FINAL CHECK IS ON THE ORIGINAL SYSTEM

Not on the substituted one. A pair that satisfies $u + v = 5$ may still fail $\sqrt{x} + \sqrt{y} = 5$ if a sign was lost. Substitute the numbers back into the equations as first written.

02 Worked examples

EXAMPLE 1 Solve $\sqrt{x} + \sqrt{y} = 7, x + y = 25$.

① $u + v = 7, u^2 + v^2 = 25, \text{ both } \geq 0.$

② $uv = \frac{49 - 25}{2} = 12$

③ $t^2 - 7t + 12 = 0 \Rightarrow t = 3, 4$

④ $x = 9, y = 16$ or the reverse.

ANSWER

$(9, 16)$ and $(16, 9)$

EXAMPLE 2 Solve $\sqrt{x} - \sqrt{y} = 1, x + y = 25$.

① $u - v = 1, u^2 + v^2 = 25$.

② $u = v + 1 \Rightarrow (v+1)^2 + v^2 = 25$

③ $2v^2 + 2v - 24 = 0 \Rightarrow v^2 + v - 12 = 0$
 $v = 3$ or $v = -4$ (rejected)

④ $v = 3, u = 4$

ANSWER

(16, 9)

EXAMPLE 3 Solve $\sqrt{x+y} = 4, x - y = 2$.

① $x + y = 16$
 Condition satisfied.

② $x - y = 2$

③ Add: $2x = 18 \Rightarrow x = 9$.

④ $y = 7$. Check $\sqrt{16} = 4$ ✓

ANSWER

(9, 7)

03 Interactive model

Write the two conditions on the board and keep them visible for the whole solution.

Substituting a root

$$\sqrt[3]{64} = 64^{1/3} = 4$$

base a **64**

index n **3**

power m **1**

check: value³ **64**

a¹ **64**

Raising the answer to the power 3 gives back a¹ — that is exactly what “root of degree 3” means.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
System of irrational equations	Irratsional tenglamalar sistemasi	Система иррациональных уравнений
Non-negativity condition	Manfiy emaslik sharti	Условие неотрицательности
Auxiliary variable	Yordamchi o‘zgaruvchi	Вспомогательная переменная
Sum of roots	Ildizlar yig‘indisi	Сумма корней
Product of roots	Ildizlar ko‘paytmasi	Произведение корней
Back-substitution	Teskari almashtirish	Обратная подстановка
Rejected pair	Rad etilgan juftlik	Отброшенная пара
Symmetric system	Simmetrik sistema	Симметричная система
Verification	Tekshirish	Проверка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

In $u = \sqrt{x}$ the condition is:

If $u = 3$ then x is:

$\sqrt{x} + \sqrt{y} = 5$, $x + y = 13$ gives uv :

A negative value of v means:

a second answer

the pair is rejected

square it

nothing

The final check is done on:

the substituted system

the original system

neither

the quadratic

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Solve $\sqrt{x} = 2, y = x + 1$

ANSWER $(4, 5)$

2. Solve $\sqrt{x + y} = 3, x = y$

ANSWER $(4.5, 4.5)$

3. If $u = \sqrt{x} = 5$, find x

ANSWER 25

4. Condition on $u = \sqrt{x}$

ANSWER $u \geq 0$

5. Solve $\sqrt{x} + \sqrt{y} = 0$

ANSWER $(0, 0)$

6. Solve $\sqrt{x} = 3, \sqrt{y} = 4$

ANSWER $(9, 16)$

7. Write $x + y$ in terms of u, v

ANSWER $u^2 + v^2$

Medium · the standard question

7 PROBLEMS

1. Solve $\sqrt{x} + \sqrt{y} = 5, x + y = 13$

ANSWER (4,9), (9,4)

2. Solve $\sqrt{x} + \sqrt{y} = 7, x + y = 25$

ANSWER (9,16), (16,9)

3. Solve $\sqrt{x} - \sqrt{y} = 1, x + y = 25$

ANSWER (16, 9)

4. Solve $\sqrt{x+y} = 4, x - y = 2$

ANSWER (9, 7)

5. Solve $\sqrt{x} + \sqrt{y} = 6, \sqrt{xy} = 8$

ANSWER (4,16), (16,4)

6. Solve $\sqrt{x} \cdot \sqrt{y} = 6, x + y = 13$

ANSWER (4,9), (9,4)

7. Solve $\sqrt{x-1} = 2, x + y = 10$

ANSWER (5, 5)

Hard · stretch and reason

7 PROBLEMS

1. Solve $\sqrt{x} + \sqrt{y} = 4, x + y = 10$

ANSWER $(3 \pm 2\sqrt{2})^2$ — i.e. $u, v = 2 \pm \sqrt{2}$

2. Solve $x + y = 20, \sqrt{x} - \sqrt{y} = 2$

ANSWER $(16, 4)$

3. Solve $\sqrt{x+y} + \sqrt{x-y} = 4, x^2 - y^2 = 9$

ANSWER $(5, 4)$

4. Solve $\frac{1}{\sqrt{x}} + \frac{1}{\sqrt{y}} = \frac{5}{6}, xy = 36$

ANSWER $(4, 9), (9, 4)$

5. Solve $\sqrt{x} + \sqrt{y} = \sqrt{x+y}$

ANSWER $x = 0$ or $y = 0$

6. Explain why $\sqrt{x} + \sqrt{y} = 3, x + y = 10$ has no solution

ANSWER $uv = -0.5 < 0$ is impossible for $u, v \geq 0$

7. Solve $\sqrt{x} + \sqrt{y} = 5, x - y = 5$

ANSWER $(9, 4)$

07 Homework

Homework — 6 tasks

Write $u \geq 0$ and $v \geq 0$ on the first line of every solution.

1. Solve $\sqrt{x} + \sqrt{y} = 9, x + y = 45$.

2. Solve $\sqrt{x} - \sqrt{y} = 2, x + y = 20$.

3. Solve $\sqrt{x+y} = 5, x - y = 3$.

4. Solve $\sqrt{x} + \sqrt{y} = 8, \sqrt{xy} = 15$.
5. Explain why $\sqrt{x} + \sqrt{y} = 2, x + y = 10$ has no solution.
6. Check both answers of task 1 by substituting into the original system.